In [100…
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix
from sklearn.model_selection import train_test_split
```

In [101…
```
df=pd.read_csv('weather-check.csv')
df
```

Out[101…

	RespondentID	Check_Weather_Daily	Check_Weather_Method	Weather_Smart
0	3887201482	Yes	The default weather app on your phone	
1	3887159451	Yes	The default weather app on your phone	
2	3887152228	Yes	The default weather app on your phone	
3	3887145426	Yes	The default weather app on your phone	
4	3887021873	Yes	A specific website or app (please provide the ...	
...	...	...	...	
914	3877568315	Yes	The Weather Channel	
915	3877568116	No	Internet search	
916	3877568054	Yes	The Weather Channel	
917	3877568053	Yes	The default weather app on your phone	
918	3877566926	Yes	Local TV News	

919 rows × 8 columns

In [102…
```
df.isnull().sum()
```

Out[102…
```
RespondentID                  0
Check_Weather_Daily           0
Check_Weather_Method          0
Weather_Smartwatch_Likelihood 0
Age                           0
Gender                        0
Household_Income              0
US Region                     0
dtype: int64
```

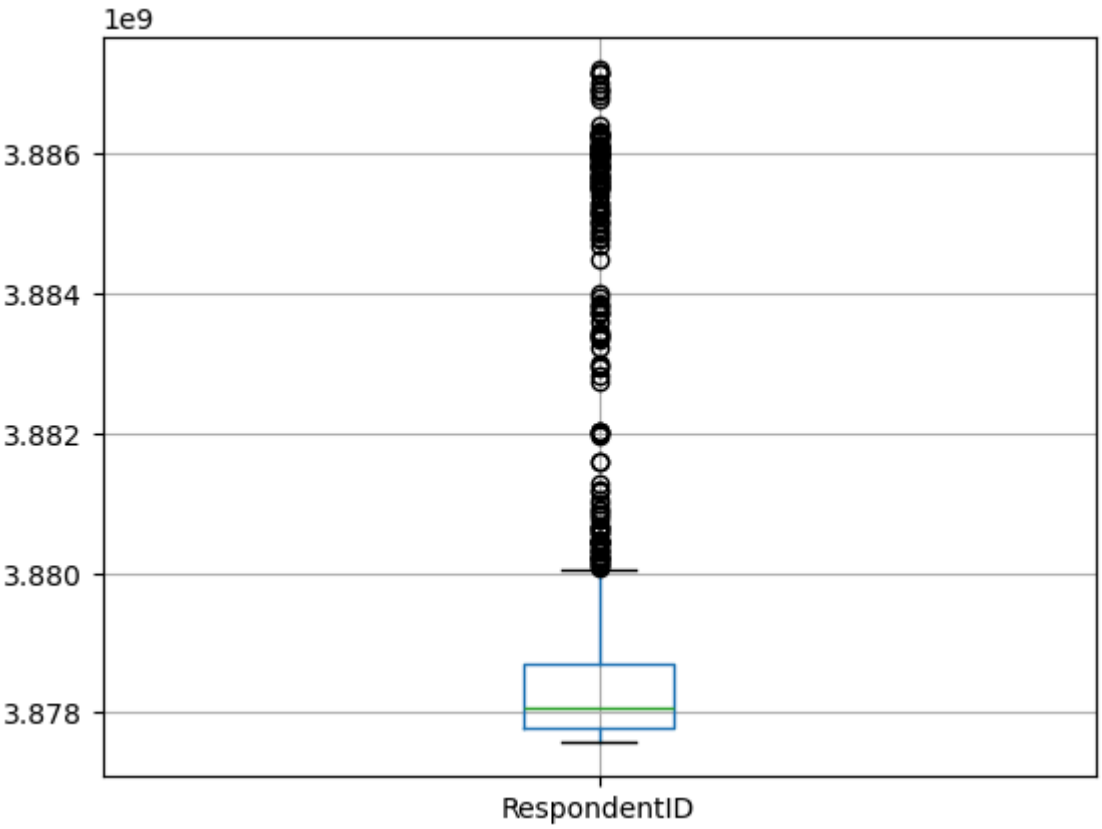
In [103…
```
df.duplicated().sum()
```

Out[103…
```
np.int64(0)
```

In [104…
```
df['Weather_Smartwatch_Likelihood'].value_counts()
```

Out[104…
```
Weather_Smartwatch_Likelihood
Very likely          362
Somewhat likely      275
Very unlikely        209
Somewhat unlikely     73
Name: count, dtype: int64
```

In [114…
```
df.select_dtypes(include='number').boxplot()
plt.show()
```



In [106…
```
df.columns
```

Out[106…
```
Index(['RespondentID', 'Check_Weather_Daily', 'Check_Weather_Method',
       'Weather_Smartwatch_Likelihood', 'Age', 'Gender', 'Household_Inco
me',
       'US Region'],
      dtype='object')
```

In [107…
```
df.shape
```

Out[107…
```
(919, 8)
```

In [108…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 919 entries, 0 to 918
Data columns (total 8 columns):
 #   Column                          Non-Null Count  Dtype 
---  ------                          --------------  ----- 
 0   RespondentID                    919 non-null    int64 
 1   Check_Weather_Daily             919 non-null    object
 2   Check_Weather_Method            919 non-null    object
 3   Weather_Smartwatch_Likelihood   919 non-null    object
 4   Age                             919 non-null    object
 5   Gender                          919 non-null    object
 6   Household_Income                919 non-null    object
 7   US Region                       919 non-null    object
dtypes: int64(1), object(7)
memory usage: 57.6+ KB
```

In [109…
```
df.describe()
```

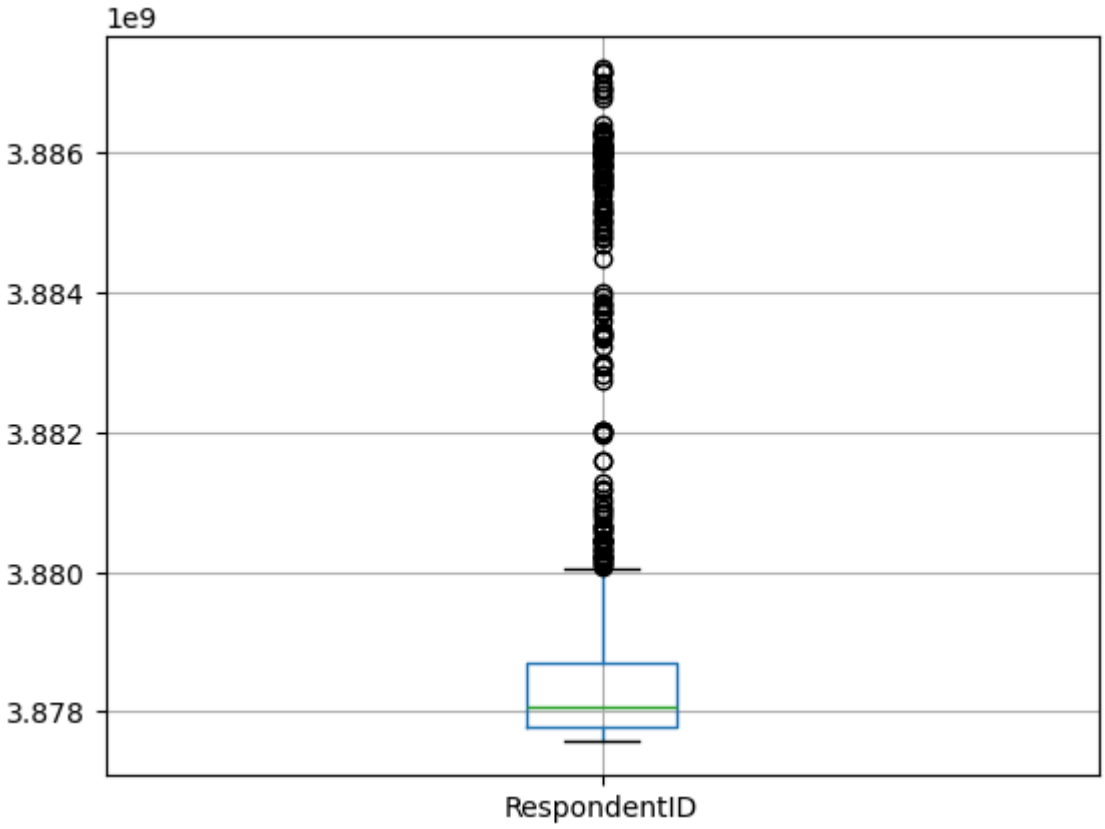
Out[109…

	RespondentID
count	9.190000e+02
mean	3.879084e+09
std	2.445434e+06
min	3.877567e+09
25%	3.877793e+09
50%	3.878059e+09
75%	3.878700e+09
max	3.887201e+09

In [110…
```
df.dtypes
```

Out[110…
```
RespondentID                     int64
Check_Weather_Daily             object
Check_Weather_Method            object
Weather_Smartwatch_Likelihood   object
Age                             object
Gender                          object
Household_Income                object
US Region                       object
dtype: object
```

In [111…
```
df.boxplot(column='RespondentID')
plt.show()
```



In [112…
```
model = DecisionTreeClassifier()
```

In [113…
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix

df = pd.read_csv("weather-check.csv")
df = df.dropna()

X = df.drop('Weather_Smartwatch_Likelihood', axis=1)
y = df['Weather_Smartwatch_Likelihood']

X = pd.get_dummies(X, drop_first=True)

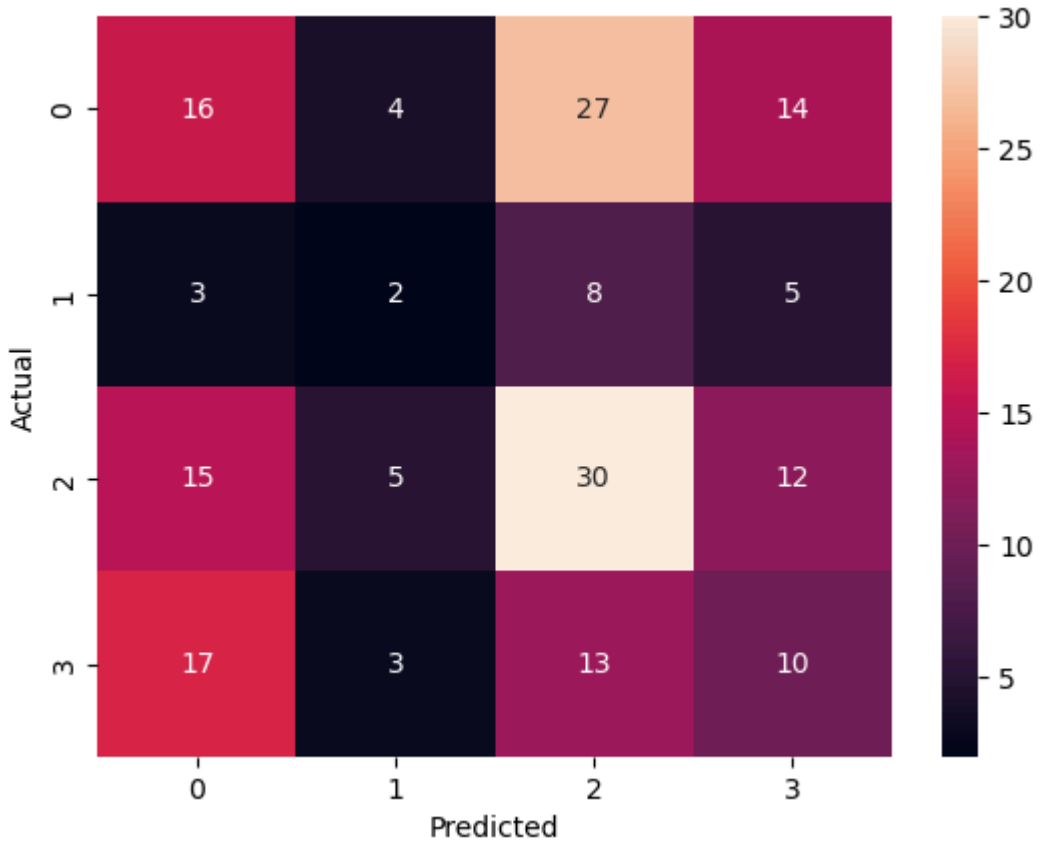
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt='d')
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.show()
```



In [115…
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix
```

In [116…
```
df = pd.read_csv("weather-check.csv")
df = df.dropna()
```

In [117…
```
X = df.drop('Weather_Smartwatch_Likelihood', axis=1)
y = df['Weather_Smartwatch_Likelihood']
```

In [118…
```
X = pd.get_dummies(X, drop_first=True)
```

In [119…
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

In [120…
```
model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)
```

Out[120…
▾ DecisionTreeClassifier

▶ Parameters

In [121…
```
y_pred = model.predict(X_test)
```

In [122…
```
confu_matrix = confusion_matrix(y_test, y_pred)
```

In [ ]: